

Problem A. Vector-Sort

OS Linux

You are given N integers. Sort the N integers and print the sorted order.

Store the N integers in a vector. Vectors are sequence containers representing arrays that can change in size.

- *Declaration:*

```
1 | vector<int>v; (creates an empty vector of integers)
```

- *Size:*

```
int size=v.size();
```

- *Pushing an integer into a vector:*

```
1 | v.push_back(x); (where x is an integer. The size increases by 1 after)
```

- *Popping the last element from the vector:*

```
1 | v.pop_back(); (After this the size decreases by 1)
```

- *Sorting a vector:*

```
1 | sort(v.begin(),v.end()); (Will sort all the elements in the vector)
```

To know more about vectors, [Click Here](#)

Input Format

The first line of the input contains N where N is the number of integers. The next line contains N integers.

Constraints

$$1 \leq N \leq 10^5$$

$$1 \leq V_i \leq 10^9, \text{ where } V_i \text{ is the } i^{\text{th}} \text{ integer in the vector.}$$

Output Format

Print the integers in the sorted order one by one in a single line followed by a space.

Input	Output
5 1 6 10 8 4	1 4 6 8 10

Problem B. Vector-Erase

OS Linux

You are provided with a vector of N integers. Then, you are given **2** queries. For the first query, you are provided with **1** integer, which denotes a position in the vector. The value at this position in the vector needs to be erased. The next query consists of **2** integers denoting a range of the positions in the vector. The elements which fall under that range should be removed. The second query is performed on the updated vector which we get after performing the first query.

The following are some useful vector functions:

- `erase(int position):`

```
1 | Removes the element present at position.
2 | Ex: v.erase(v.begin()+4); (erases the fifth element of the vector)
```

- `erase(int start,int end):`

```
1 | Removes the elements in the range from start to end inclusive of
2 | Ex:v.erase(v.begin()+2,v.begin()+5);(erases all the elements from)
```

Input Format

The first line of the input contains an integer N .The next line contains N space separated integers(1-based index).The third line contains a single integer x ,denoting the position of an element that should be removed from the vector.The fourth line contains two integers a and b denoting the range that should be erased from the vector inclusive of a and exclusive of b .

Constraints

$$1 \leq N \leq 10^5$$

$$1 \leq x \leq N$$

$$1 \leq a < b \leq N$$

Output Format

Print the size of the vector in the first line and the elements of the vector after the two erase operations in the second line separated by space.

Input	Output
6 1 4 6 2 8 9 2 2 4	3 1 8 9

Explanation

The first query is to erase the 2nd element in the vector, which is 4. Then, modified vector is {1 6 2 8 9}, we want to remove the range of 2~4, which means the 2nd and 3rd elements should be removed. Then 6 and 2 in the modified vector are removed and we finally get {1 8 9}

Problem C. Swappable

Time Limit 2000 ms

Problem Statement

Given an array of N integers $A = (A_1, A_2, \dots, A_N)$, find the number of pairs (i, j) of integers satisfying all of the following conditions:

- $1 \leq i < j \leq N$
- $A_i \neq A_j$

Constraints

- All values in input are integers.
- $2 \leq N \leq 3 \times 10^5$
- $1 \leq A_i \leq 10^9$

Input

Input is given from Standard Input in the following format:

```
N
A1 A2 ... AN
```

Output

Print the answer as an integer.

Sample 1

Input	Output
3 1 7 1	2

In this input, we have $A = (1, 7, 1)$.

- For the pair $(1, 2)$, $A_1 \neq A_2$.

- For the pair $(1, 3)$, $A_1 = A_3$.
- For the pair $(2, 3)$, $A_2 \neq A_3$.

Sample 2

Input	Output
10 1 10 100 1000 10000 100000 1000000 10000000 100000000 1000000000	45

Sample 3

Input	Output
20 7 8 1 1 4 9 9 6 8 2 4 1 1 9 5 5 5 3 6 4	173

Problem D. Sets-STL

OS Linux

Sets are a part of the C++ STL. Sets are containers that store unique elements following a specific order. Here are some of the frequently used member functions of sets:

- *Declaration:*

```
1 | set<int>s; //Creates a set of integers.
```

- *Size:*

```
1 | int length=s.size(); //Gives the size of the set.
```

- *Insert:*

```
1 | s.insert(x); //Inserts an integer x into the set s.
```

- *Erasing an element:*

```
1 | s.erase(val); //Erases an integer val from the set s.
```

- *Finding an element:*

```
1 | set<int>::iterator itr=s.find(val); //Gives the iterator to the element
2 | Ex: set<int>::iterator itr=s.find(100); //If 100 is not present t
```

To know more about sets [click Here](#). Coming to the problem, you will be given Q queries. Each query is of one of the following three types:

1 x : Add an element x to the set.

2 x : Delete an element x from the set. (If the number x is not present in the set, then do nothing).

3 x : If the number x is present in the set, then print "Yes"(without quotes) else print "No"(without quotes).

Input Format

The first line of the input contains Q where Q is the number of queries. The next Q lines contain 1 query each. Each query consists of two integers y and x where y is the type of the query and x is an integer.

Constraints

$$1 \leq Q \leq 10^5$$

$$1 \leq y \leq 3$$

$$1 \leq x \leq 10^9$$

Output Format

For queries of type 3 print "Yes"(without quotes) if the number x is present in the set and if the number is not present, then print "No"(without quotes).

Each query of type 3 should be printed in a new line.

Input	Output
8	Yes
1 9	No
1 6	No
1 10	
1 4	
3 6	
3 14	
2 6	
3 6	

Problem E. 2D-SORT

Time Limit	100 ms
Mem Limit	1572864 kB
Code Length Limit	50000 B
OS	Linux

Given n points in a two dimensional space, sort all the points in ascending order.

$(x_1, y_1) > (x_2, y_2)$ if and only if $(x_1 > x_2)$ or $(x_1 == x_2 \text{ and } y_1 < y_2)$

Input

The first line consists of an integer t , the number of test cases. Then for each test case the first line consists of an integer n , the number of points. Then the next n lines contains two integers x_i, y_i which represents the point.

Output

For each test case print the sorted order of the points.

Constraints

$$1 \leq t \leq 10$$

$$1 \leq n \leq 100000$$

$$-10^9 \leq \text{coordinates} \leq 10^9$$

NOTE: Strict time limit. Prefer scanf/printf/BufferedReader instead of cin/cout/Scanner.

Example

Input	Output
1 5 3 4 -1 2 5 -3 3 3 -1 -2	-1 2 -1 -2 3 4 3 3 5 -3

Problem F. FIGUREFUL

Time Limit	1000 ms
Mem Limit	1572864 kB
Code Length Limit	50000 B
OS	Linux

Don and his friends had some sort of code word to describe the girls who pass towards their class.

- Anitha passed – killer told.. “Dude, 5 2 is going”
- Chitra passed – Jumbo told.. “Dude, 6 4 is awesome”
- Divya passed – Baabi told.. “Machi 1 6 is my dream girl”

Girls couldn't understand those code words. To crack the code word, girls decide to record all the code words and the girls' name whenever they call.

The silencer in the class watched all these and got angry. He mugged up all the code words, went to teacher and complained about the naughty boys and naughty girls.

- **Teacher:** Who is the girl with code word 1 6?
- **Silencer:** Divya
- **Teacher:** Who is the girl with code word 5 2?
- **Silencer:** Anitha
- **Teacher:** What is the code word for me?
- **Silencer:** (puzzled)

Now you have to pretend that you are the Silencer to solve this problem. Record the code words and answer the questions the teacher asks you. (She may ask the same question any number of times.)

Note: There can be any number of code words for a same girl. If there exist more than one girl for a same code word, then that code word belongs to the latest girl.

Input

The first line contains an integer n , the number of figures. Then in each of the next n lines there will be 2 integers and a figure name.

After that there will be an integer t , the number of test cases. For each test case, there will be two integers representing the code word. Print the figure's name corresponding to the code word.

Output

For each test case output the figure name.

Constraints

$1 \leq n \leq 100000$

$1 \leq t \leq 100000$

$-1000000 \leq \text{code1}, \text{code2} \leq 1000000$

The figure's name will not exceed 20 characters.

Example

Input	Output
6 5 2 Anitha 6 4 Yamuna 6 4 Chitra 1 6 Divya -2 0 Teacher 7 4 Anitha 2 -2 0 6 4	Teacher Chitra

Hint: Use map.

Problem G. Maps-STL

OS Linux

Maps are a part of the C++ STL. Maps are associative containers that store elements formed by a combination of a key value and a mapped value, following a specific order. The mainly used member functions of maps are:

- *Map Template:*

```
1 | std::map <key_type, data_type>
```

- *Declaration:*

```
1 | map<string,int>m; //Creates a map m where key_type is of type str
```

- *Size:*

```
1 | int length=m.size(); //Gives the size of the map.
```

- *Insert:*

```
1 | m.insert(make_pair("hello",9)); //Here the pair is inserted into
```

- *Erasing an element:*

```
1 | m.erase(val); //Erases the pair from the map where the key_type i
```

- *Finding an element:*

```
1 | map<string,int>::iterator itr=m.find(val); //Gives the iterator t
2 | Ex: map<string,int>::iterator itr=m.find("Maps"); //If Maps is no
```

- *Accessing the value stored in the key:*

```
1 | To get the value stored of the key "MAPS" we can do m["MAPS"] or
```

To know more about maps [click Here](#).

You are appointed as the assistant to a teacher in a school and she is correcting the answer sheets of the students. Each student can have multiple answer sheets. So the teacher has Q queries:

1 X Y : Add the marks Y to the student whose name is X .

2 X : Erase the marks of the students whose name is X .

3 X : Print the marks of the students whose name is X . (If X didn't get any marks print 0.)

Input Format

The first line of the input contains Q where Q is the number of queries. The next Q lines contain 1 query each. The first integer, *type* of each query is the type of the query. If query is of type 1, it consists of one string and an integer X and Y where X is the name of the student and Y is the marks of the student. If query is of type 2 or 3, it consists of a single string X where X is the name of the student.

Constraints

$$1 \leq Q \leq 10^5$$

$$1 \leq \textit{type} \leq 3$$

$$1 \leq |X| \leq 6$$

$$1 \leq Y \leq 10^3$$

Output Format

For queries of type 3 print the marks of the given student.

Input	Output
7 1 Jesse 20 1 Jess 12 1 Jess 18 3 Jess 3 Jesse 2 Jess 3 Jess	30 20 0

Problem H. Koxia and Whiteboards

Time Limit 1000 ms

Mem Limit 262144 kB

Kiyora has n whiteboards numbered from 1 to n . Initially, the i -th whiteboard has the integer a_i written on it.

Koxia performs m operations. The j -th operation is to choose one of the whiteboards and change the integer written on it to b_j .

Find the maximum possible sum of integers written on the whiteboards after performing all m operations.

Input

Each test consists of multiple test cases. The first line contains a single integer t ($1 \leq t \leq 1000$) — the number of test cases. The description of test cases follows.

The first line of each test case contains two integers n and m ($1 \leq n, m \leq 100$).

The second line of each test case contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 10^9$).

The third line of each test case contains m integers $b_1, b_2, \dots, b_m$ ($1 \leq b_i \leq 10^9$).

Output

For each test case, output a single integer — the maximum possible sum of integers written on whiteboards after performing all m operations.

Examples

Input	Output
4 3 2 1 2 3 4 5 2 3 1 2 3 4 5 1 1 100 1 5 3 1 1 1 1 1 1000000000 1000000000 1000000000	12 9 1 3000000002

Note

In the first test case, Koxia can perform the operations as follows:

1. Choose the 1-st whiteboard and rewrite the integer written on it to $b_1 = 4$.
2. Choose the 2-nd whiteboard and rewrite to $b_2 = 5$.

After performing all operations, the numbers on the three whiteboards are 4, 5 and 3 respectively, and their sum is 12. It can be proven that this is the maximum possible sum achievable.

In the second test case, Koxia can perform the operations as follows:

1. Choose the 2-nd whiteboard and rewrite to $b_1 = 3$.
2. Choose the 1-st whiteboard and rewrite to $b_2 = 4$.
3. Choose the 2-nd whiteboard and rewrite to $b_3 = 5$.

The sum is $4 + 5 = 9$. It can be proven that this is the maximum possible sum achievable.

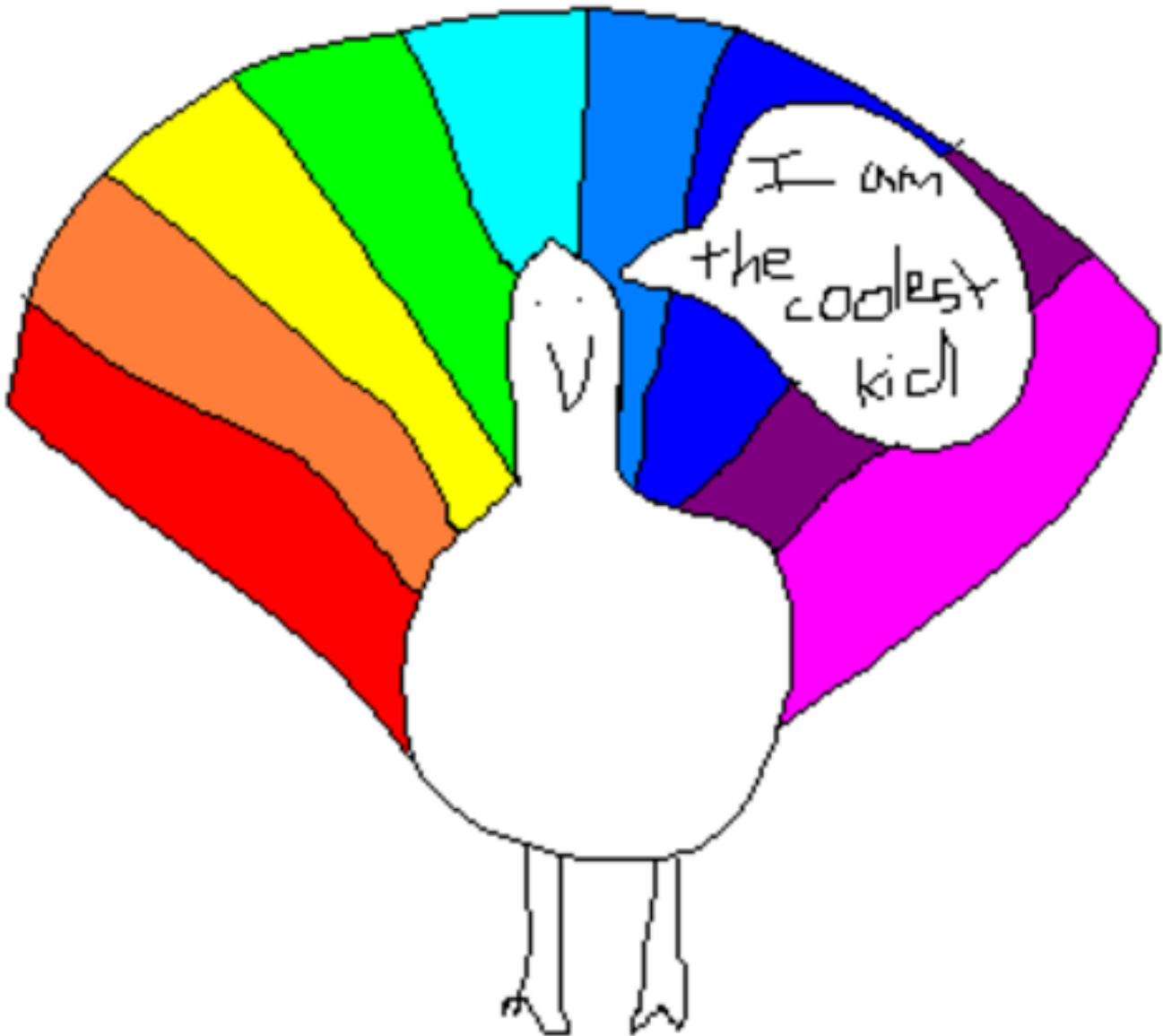
Problem I. Peacock Party

Time Limit 1500 ms

Mem Limit 262144 kB

OS Windows

Percy the peacock is quite the fancy peacock, and loves to go to peacock parties! For each party, he adorns a set of colorful feathers to look his best.



However, he recently realized that despite owning many different colors, Percy only wears a handful of colors. The problem is, Percy leaves all his colorful feathers in a pile – taking the top one off when he attends a party, and putting it back on top when he's done. In fact, the only time the color he wears ever changes is when he buys a new set of feathers!

Instead of taking from the top of the pile, Percy has now decided to take his next set of feathers from the bottom of the pile, in the hopes that he can go show off all his colors at the peacock parties. Can you help him figure out what colors he will wear to each party?

Input

The first line of input is a single integer, $1 \leq n \leq 2 \cdot 10^5$, denoting the number of days.

Then, follow n lines, where the i th line contains what Percy does on the i th day. This will be one of:

- 1 c : Percy buys a set of feathers with color c , which is a string of at most 10 characters. These feathers go on the top of the pile.
- 2: Percy attends a party with the bottommost set of feathers, and puts them back on top of the pile after he comes back from the party. It is guaranteed that the pile is not empty when this happens.

Output

For each day when Percy attends a party, output the color of the feathers he wears to the party.

Examples

Input	Output
10 1 red 1 blue 1 green 2 2 1 yellow 2 2 2 2	red blue green red blue yellow

Problem J. Balanced Brackets

OS Linux

A bracket is considered to be any one of the following characters: `(`, `)`, `{`, `}`, `[`, or `]`.

Two brackets are considered to be a *matched pair* if the an opening bracket (i.e., `(`, `[`, or `{`) occurs to the left of a closing bracket (i.e., `)`, `]`, or `}`) of the *exact same type*. There are three types of matched pairs of brackets: `[]`, `{}`, and `()`.

A matching pair of brackets is *not balanced* if the set of brackets it encloses are not matched. For example, `{[(())]}` is not balanced because the contents in between `{` and `}` are not balanced. The pair of square brackets encloses a single, unbalanced opening bracket, `(`, and the pair of parentheses encloses a single, unbalanced closing square bracket, `]`.

By this logic, we say a sequence of brackets is *balanced* if the following conditions are met:

- It contains no unmatched brackets.
- The subset of brackets enclosed within the confines of a matched pair of brackets is also a matched pair of brackets.

Given n strings of brackets, determine whether each sequence of brackets is balanced. If a string is balanced, return `YES`. Otherwise, return `NO`.

Function Description

Complete the function `isBalanced` in the editor below.

`isBalanced` has the following parameter(s):

- *string s*: a string of brackets

Returns

- *string*: either `YES` or `NO`

Input Format

The first line contains a single integer n , the number of strings.

Each of the next n lines contains a single string s , a sequence of brackets.

Constraints

- $1 \leq n \leq 10^3$
- $1 \leq |s| \leq 10^3$, where $|s|$ is the length of the sequence.

- All characters in the sequences $\in \{ \{, \}, (,), [,] \}$.

Output Format

For each string, return **YES** or **NO**.

Input		Output
STDIN	Function	YES
-----	-----	NO
3	n = 3	YES
{[()]}	first s = '[()]'	
{[()]}	second s = '[()]'	
{{[[[()]]]}}	third s = '{{[[[()]]]}'	

Explanation

- The string **{[()]}** meets both criteria for being a balanced string.
- The string **{[()]}** is not balanced because the brackets enclosed by the matched pair **{** and **}** are not balanced: **[()]**.
- The string **{{[[[()]]]}}** meets both criteria for being a balanced string.

Problem K. Text Reverse

Time Limit 1000 ms

Mem Limit 32768 kB

OS Windows

Ignatius likes to write words in reverse way. Given a single line of text which is written by Ignatius, you should reverse all the words and then output them.

Input

The input contains several test cases. The first line of the input is a single integer T which is the number of test cases. T test cases follow.

Each test case contains a single line with several words. There will be at most 1000 characters in a line.

Output

For each test case, you should output the text which is processed.

Sample

Input	Output
3 olleh !dlrow m'I morf .udh I ekil .mca	hello world! I'm from hdu. I like acm.

Hint

Remember to use `getchar()` to read `'\n'` after the integer T , then you may use `gets()` to read a line and process it.

Problem L. ACboy needs your help again!

Time Limit 1000 ms

Mem Limit 32768 kB

OS Windows

ACboy was kidnapped!!

he miss his mother very much and is very scare now.You can't image how dark the room he was put into is, so poor :(.

As a smart ACMer, you want to get ACboy out of the monster's labyrinth.But when you arrive at the gate of the maze, the monste say : " I have heard that you are very clever, but if can't solve my problems, you will die with ACboy."

The problems of the monster is shown on the wall:

Each problem's first line is a integer N(the number of commands), and a word "FIFO" or "FILO".(you are very happy because you know "FIFO" stands for "First In First Out", and "FILO" means "First In Last Out").

and the following N lines, each line is "IN M" or "OUT", (M represent a integer).

and the answer of a problem is a passowrd of a door, so if you want to rescue ACboy, answer the problem carefully!

Input

The input contains multiple test cases.

The first line has one integer,represent the number of test cases.

And the input of each subproblem are described above.

Output

For each command "OUT", you should output a integer depend on the word is "FIFO" or "FILO", or a word "None" if you don't have any integer.

Sample

Input	Output
4 4 FIFO IN 1 IN 2 OUT OUT 4 FILO IN 1 IN 2 OUT OUT 5 FIFO IN 1 IN 2 OUT OUT OUT 5 FILO IN 1 IN 2 OUT IN 3 OUT	1 2 2 1 1 2 None 2 3

Problem M. Epic Transformation

Time Limit 2000 ms

Mem Limit 262144 kB

You are given an array a of length n consisting of integers. You can apply the following operation, consisting of several steps, on the array a **zero** or more times:

- you select two **different** numbers in the array a_i and a_j ;
- you remove i -th and j -th elements from the array.

For example, if $n = 6$ and $a = [1, 6, 1, 1, 4, 4]$, then you can perform the following sequence of operations:

- select $i = 1, j = 5$. The array a becomes equal to $[6, 1, 1, 4]$;
- select $i = 1, j = 2$. The array a becomes equal to $[1, 4]$.

What can be the minimum size of the array after applying some sequence of operations to it?

Input

The first line contains a single integer t ($1 \leq t \leq 10^4$). Then t test cases follow.

The first line of each test case contains a single integer n ($1 \leq n \leq 2 \cdot 10^5$) is length of the array a .

The second line of each test case contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 10^9$).

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output the minimum possible size of the array after applying some sequence of operations to it.

Examples

Input	Output
5	0
6	0
1 6 1 1 4 4	2
2	1
1 2	0
2	
1 1	
5	
4 5 4 5 4	
6	
2 3 2 1 3 1	

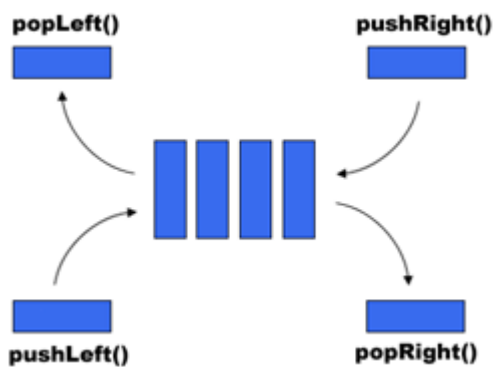
Problem N. Double Ended Queue

Time Limit 1000 ms

Mem Limit 65536 kB

A queue is a data structure based on the principle of 'First In First Out' (FIFO). There are two ends; one end can be used only to insert an item and the other end to remove an item. A Double Ended Queue is a queue where you can insert an item in both sides as well as you can delete an item from either side.

There are mainly four operations available to a double ended queue. They are:



1. `pushLeft()` inserts an item to the left end of the queue with the exception that the queue is not full.
2. `pushRight()` inserts an item to the right end of the queue with the exception that the queue is not full.
3. `popLeft()` removes an item from the left end of the queue with the exception that the queue is not empty.
4. `popRight()` removes an item from the right end of the queue with the exception that the queue is not empty.

Now you are given a queue and a list of commands, you have to report the behavior of the queue.

Input

Input starts with an integer T (≤ 20), denoting the number of test cases. Each case starts with a line containing two integers n, m ($1 \leq n \leq 10, 1 \leq m \leq 100$), where n denotes the size of the queue and m denotes the number of commands. Each of the next m lines contains a command which is one of:

Operation	Action
pushLeft x	pushes x ($-100 \leq x \leq 100$) to the left end of the queue
pushRight x	pushes x ($-100 \leq x \leq 100$) to the right end of the queue
popLeft	pops an item from the left end of the queue
popRight	pops an item from the right end of the queue

Output

For each case, print the case number in a line. Then for each operation, show its corresponding output as shown in the sample. Be careful about spelling.

Sample

Input	Output
1 3 8 pushLeft 1 pushLeft 2 pushRight -1 pushRight 1 popLeft popRight popLeft popRight	Case 1: Pushed in left: 1 Pushed in left: 2 Pushed in right: -1 The queue is full Popped from left: 2 Popped from right: -1 Popped from left: 1 The queue is empty

Problem O. Soldier and Cards

Time Limit 2000 ms

Mem Limit 262144 kB

Two bored soldiers are playing card war. Their card deck consists of exactly n cards, numbered from 1 to n , **all values are different**. They divide cards between them in some manner, it's possible that they have different number of cards. Then they play a "war"-like card game.

The rules are following. On each turn a *fight* happens. Each of them picks card from the top of his stack and puts on the table. The one whose card value is bigger wins this *fight* and takes both cards from the table to the bottom of his stack. More precisely, he first takes his opponent's card and puts to the bottom of his stack, and then he puts his card to the bottom of his stack. If after some turn one of the player's stack becomes empty, he loses and the other one wins.

You have to calculate how many *fights* will happen and who will win the game, or state that game won't end.

Input

First line contains a single integer n ($2 \leq n \leq 10$), the number of cards.

Second line contains integer k_1 ($1 \leq k_1 \leq n - 1$), the number of the first soldier's cards. Then follow k_1 integers that are the values on the first soldier's cards, from top to bottom of his stack.

Third line contains integer k_2 ($k_1 + k_2 = n$), the number of the second soldier's cards. Then follow k_2 integers that are the values on the second soldier's cards, from top to bottom of his stack.

All card values are different.

Output

If somebody wins in this game, print 2 integers where the first one stands for the number of *fights* before end of game and the second one is 1 or 2 showing which player has won.

If the game won't end and will continue forever output - 1.

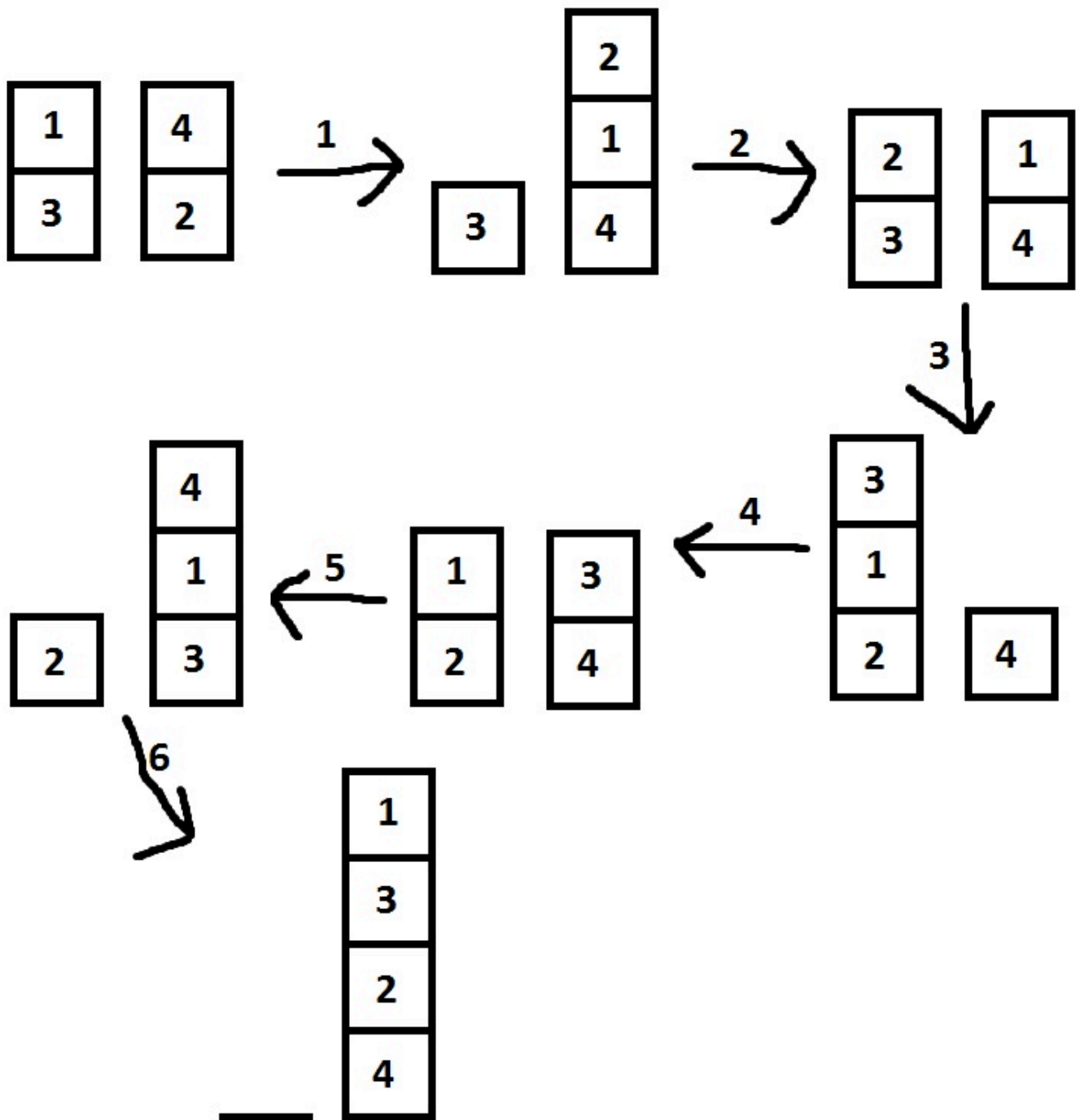
Examples

Input	Output
4 2 1 3 2 4 2	6 2

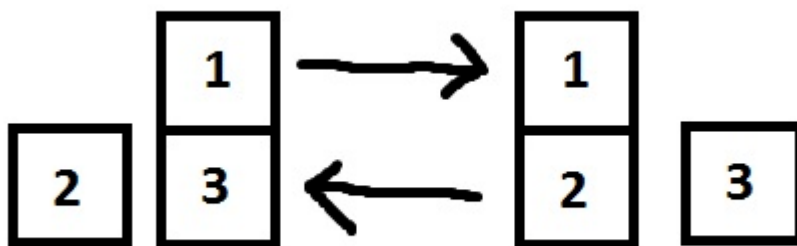
Input	Output
3 1 2 2 1 3	- 1

Note

First sample:



Second sample:



Problem P. Misha and Changing Handles

Time Limit 1000 ms

Mem Limit 262144 kB

Input File stdin

Output File stdout

Misha hacked the Codeforces site. Then he decided to let all the users change their handles. A user can now change his handle any number of times. But each new handle must not be equal to any handle that is already used or that was used at some point.

Misha has a list of handle change requests. After completing the requests he wants to understand the relation between the original and the new handles of the users. Help him to do that.

Input

The first line contains integer q ($1 \leq q \leq 1000$), the number of handle change requests.

Next q lines contain the descriptions of the requests, one per line.

Each query consists of two non-empty strings *old* and *new*, separated by a space. The strings consist of lowercase and uppercase Latin letters and digits. Strings *old* and *new* are distinct. The lengths of the strings do not exceed 20.

The requests are given chronologically. In other words, by the moment of a query there is a single person with handle *old*, and handle *new* is not used and has not been used by anyone.

Output

In the first line output the integer n — the number of users that changed their handles at least once.

In the next n lines print the mapping between the old and the new handles of the users. Each of them must contain two strings, *old* and *new*, separated by a space, meaning that before the user had handle *old*, and after all the requests are completed, his handle is *new*. You may output lines in any order.

Each user who changes the handle must occur exactly once in this description.

Examples

Input	Output
5 Misha ILoveCodeforces Vasya Petrov Petrov VasyaPetrov123 ILoveCodeforces MikeMirzayanov Petya Ivanov	3 Petya Ivanov Misha MikeMirzayanov Vasya VasyaPetrov123

Problem Q. Lala Land and Apple Trees

Time Limit 1000 ms

Mem Limit 262144 kB

Amr lives in Lala Land. Lala Land is a very beautiful country that is located on a coordinate line. Lala Land is famous with its apple trees growing everywhere.

Lala Land has exactly n apple trees. Tree number i is located in a position x_i and has a_i apples growing on it. Amr wants to collect apples from the apple trees. Amr currently stands in $x = 0$ position. At the beginning, he can choose whether to go right or left. He'll continue in his direction until he meets an apple tree he didn't visit before. He'll take all of its apples and then reverse his direction, continue walking in this direction until he meets another apple tree he didn't visit before and so on. In the other words, Amr reverses his direction when visiting each new apple tree. Amr will stop collecting apples when there are no more trees he didn't visit in the direction he is facing.

What is the maximum number of apples he can collect?

Input

The first line contains one number n ($1 \leq n \leq 100$), the number of apple trees in Lala Land.

The following n lines contains two integers each x_i, a_i ($-10^5 \leq x_i \leq 10^5, x_i \neq 0, 1 \leq a_i \leq 10^5$), representing the position of the i -th tree and number of apples on it.

It's guaranteed that there is at most one apple tree at each coordinate. It's guaranteed that no tree grows in point 0.

Output

Output the maximum number of apples Amr can collect.

Examples

Input	Output
2 -1 5 1 5	10

Input	Output
3 -2 2 1 4 -1 3	9

Input	Output
3 1 9 3 5 7 10	9

Note

In the first sample test it doesn't matter if Amr chose at first to go left or right. In both cases he'll get all the apples.

In the second sample test the optimal solution is to go left to $x = -1$, collect apples from there, then the direction will be reversed, Amr has to go to $x = 1$, collect apples from there, then the direction will be reversed and Amr goes to the final tree $x = -2$.

In the third sample test the optimal solution is to go right to $x = 1$, collect apples from there, then the direction will be reversed and Amr will not be able to collect anymore apples because there are no apple trees to his left.

Problem R. Tom Riddle's Diary

Time Limit 2000 ms

Mem Limit 262144 kB

Harry Potter is on a mission to destroy You-Know-Who's Horcruxes. The first Horcrux that he encountered in the Chamber of Secrets is Tom Riddle's diary. The diary was with Ginny and it forced her to open the Chamber of Secrets. Harry wants to know the different people who had ever possessed the diary to make sure they are not under its influence.

He has names of n people who possessed the diary in order. You need to tell, for each person, if he/she possessed the diary at some point before or not.

Formally, for a name S_i in the i -th line, output "YES" (without quotes) if there exists an index j such that $S_i = S_j$ and $j < i$, otherwise, output "NO" (without quotes).

Input

First line of input contains an integer n ($1 \leq n \leq 100$) — the number of names in the list.

Next n lines each contain a string S_i , consisting of lowercase English letters. The length of each string is between 1 and 100.

Output

Output n lines each containing either "YES" or "NO" (without quotes), depending on whether this string was already present in the stream or not.

You can print each letter in any case (upper or lower).

Examples

Input	Output
6 tom lucius ginny harry ginny harry	NO NO NO NO YES YES YES

Input	Output
3 a a a	NO YES YES

Note

In test case 1, for $i = 5$ there exists $j = 3$ such that $s_i = s_j$ and $j < i$, which means that answer for $i = 5$ is "YES".

Problem S. A problem of sorting

Time Limit 1000 ms

Mem Limit 65536 kB

OS Windows

There are many people's name and birth in a list. Your task is to print the name from young to old. (There is no pair of two has the same age.)

Input

First line contains a single integer $T \leq 100$ which denotes the number of test cases.

For each test case, there is an positive integer n ($1 \leq n \leq 100$) which denotes the number of people, and next n lines, each line has a name and a birth's year (1900–2015) separated by one space.

The length of name is positive and not larger than 100. Notice name only contain letter(s), digit(s) and space(s).

Output

For each case, output n lines.

Sample

Input	Output
2 1 FancyCoder 1996 2 FancyCoder 1996 xyz111 1997	FancyCoder xyz111 FancyCoder

Problem T. Cutting Woods

Time Limit 2000 ms

Problem Statement

We have a long piece of timber with a length of L meters.

For each $x = 1, 2, \dots, L - 1$, there is a mark called Mark x at x meters from the left end of the piece.

You are given Q queries, the i -th of which is represented as a pair of numbers (c_i, x_i) .

Process the queries in ascending order of i as described below.

- If $c_i = 1$: cut the piece at Mark x_i into two.
- If $c_i = 2$: choose the piece with Mark x_i on it and print its length.

Here, for both kinds of queries $c_i = 1, 2$, it is guaranteed that there will have been no cut at Mark x_i when the query is to be processed.

Constraints

- $1 \leq L \leq 10^9$
- $1 \leq Q \leq 2 \times 10^5$
- $c_i = 1, 2$ ($1 \leq i \leq Q$)
- $1 \leq x_i \leq L - 1$ ($1 \leq i \leq Q$)
- For every i ($1 \leq i \leq Q$), the following holds: there is no j such that $1 \leq j < i$ and $(c_j, x_j) = (1, x_i)$.
- All values in input are integers.

Input

Input is given from Standard Input in the following format:

```
L Q
c1 x1
c2 x2
⋮
cQ xQ
```

Output

Print the number of lines equal to the number of queries $c_i = 2$. In the j -th line, print the response to the j -th such query.

Sample 1

Input	Output
5 3 2 2 1 3 2 2	5 3

At the time of the first query, no cut has been made, so the piece with Mark 2 has a length of 5 meters. Thus, you should print 5.

In the second query, the piece is cut into two pieces with lengths of 3 and 2 meters.

At the time of the third query, the piece with Mark 2 has a length of 3 meters, so you should print 3.

Sample 2

Input	Output
5 3 1 2 1 4 2 3	2

Sample 3

Input	Output
100 10 1 31 2 41 1 59 2 26 1 53 2 58 1 97 2 93 1 23 2 84	69 31 6 38 38

Problem U. Good Sequence

Time Limit 2000 ms

Problem Statement

You are given a sequence of positive integers of length N , $a = (a_1, a_2, \dots, a_N)$. Your objective is to remove some of the elements in a so that a will be a **good sequence**.

Here, an sequence b is a **good sequence** when the following condition holds true:

- For each element x in b , the value x occurs exactly x times in b .

For example, $(3, 3, 3)$, $(4, 2, 4, 1, 4, 2, 4)$ and $()$ (an empty sequence) are good sequences, while $(3, 3, 3, 3)$ and $(2, 4, 1, 4, 2)$ are not.

Find the minimum number of elements that needs to be removed so that a will be a good sequence.

Constraints

- $1 \leq N \leq 10^5$
- a_i is an integer.
- $1 \leq a_i \leq 10^9$

Input

Input is given from Standard Input in the following format:

```
N
a1 a2 ... aN
```

Output

Print the minimum number of elements that needs to be removed so that a will be a good sequence.

Sample 1

Input	Output
4 3 3 3 3	1

We can, for example, remove one occurrence of 3. Then, (3, 3, 3) is a good sequence.

Sample 2

Input	Output
5 2 4 1 4 2	2

We can, for example, remove two occurrences of 4. Then, (2, 1, 2) is a good sequence.

Sample 3

Input	Output
6 1 2 2 3 3 3	0

Sample 4

Input	Output
1 1000000000	1

Remove one occurrence of 10^9 . Then, () is a good sequence.

Sample 5

Input	Output
8 2 7 1 8 2 8 1 8	5

Problem V. Registration System

Time Limit 5000 ms

Mem Limit 65536 kB

Input File stdin

Output File stdout

A new e-mail service "Berlandesk" is going to be opened in Berland in the near future. The site administration wants to launch their project as soon as possible, that's why they ask you to help. You're suggested to implement the prototype of site registration system. The system should work on the following principle.

Each time a new user wants to register, he sends to the system a request with his name. If such a name does not exist in the system database, it is inserted into the database, and the user gets the response OK, confirming the successful registration. If the name already exists in the system database, the system makes up a new user name, sends it to the user as a prompt and *also inserts the prompt into the database*. The new name is formed by the following rule. Numbers, starting with 1, are appended one after another to name (name1, name2, ...), among these numbers the least i is found so that name i does not yet exist in the database.

Input

The first line contains number n ($1 \leq n \leq 10^5$). The following n lines contain the requests to the system. Each request is a non-empty line, and consists of not more than 32 characters, which are all lowercase Latin letters.

Output

Print n lines, which are system responses to the requests: OK in case of successful registration, or a prompt with a new name, if the requested name is already taken.

Examples

Input	Output
4 abacaba acaba abacaba acab	OK OK abacaba1 OK

Input	Output
6 first first second second third third	OK first1 OK second1 OK third1

Problem W. Worms

Time Limit 1000 ms

Mem Limit 262144 kB

Input File stdin

Output File stdout

It is lunch time for Mole. His friend, Marmot, prepared him a nice game for lunch.

Marmot brought Mole n ordered piles of worms such that i -th pile contains a_i worms. He labeled all these worms with consecutive integers: worms in first pile are labeled with numbers 1 to a_1 , worms in second pile are labeled with numbers $a_1 + 1$ to $a_1 + a_2$ and so on. See the example for a better understanding.

Mole can't eat all the worms (Marmot brought a lot) and, as we all know, Mole is blind, so Marmot tells him the labels of the best juicy worms. Marmot will only give Mole a worm if Mole says correctly in which pile this worm is contained.

Poor Mole asks for your help. For all juicy worms said by Marmot, tell Mole the correct answers.

Input

The first line contains a single integer n ($1 \leq n \leq 10^5$), the number of piles.

The second line contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 10^3$, $a_1 + a_2 + \dots + a_n \leq 10^6$), where a_i is the number of worms in the i -th pile.

The third line contains single integer m ($1 \leq m \leq 10^5$), the number of juicy worms said by Marmot.

The fourth line contains m integers $q_1, q_2, \dots, q_m$ ($1 \leq q_i \leq a_1 + a_2 + \dots + a_n$), the labels of the juicy worms.

Output

Print m lines to the standard output. The i -th line should contain an integer, representing the number of the pile where the worm labeled with the number q_i is.

Examples

Input	Output
5 2 7 3 4 9 3 1 25 11	1 5 3

Note

For the sample input:

- The worms with labels from $[1, 2]$ are in the first pile.
- The worms with labels from $[3, 9]$ are in the second pile.
- The worms with labels from $[10, 12]$ are in the third pile.
- The worms with labels from $[13, 16]$ are in the fourth pile.
- The worms with labels from $[17, 25]$ are in the fifth pile.

Problem X. Queries about less or equal elements

Time Limit 2000 ms

Mem Limit 262144 kB

You are given two arrays of integers a and b . For each element of the second array b_j you should find the number of elements in array a that are less than or equal to the value b_j .

Input

The first line contains two integers n, m ($1 \leq n, m \leq 2 \cdot 10^5$) — the sizes of arrays a and b .

The second line contains n integers — the elements of array a ($-10^9 \leq a_i \leq 10^9$).

The third line contains m integers — the elements of array b ($-10^9 \leq b_j \leq 10^9$).

Output

Print m integers, separated by spaces: the j -th of which is equal to the number of such elements in array a that are less than or equal to the value b_j .

Examples

Input	Output
5 4 1 3 5 7 9 6 4 2 8	3 2 1 4
Input	Output
5 5 1 2 1 2 5 3 1 4 1 5	4 2 4 2 5

Problem Y. T-shirt buying

Time Limit 3000 ms

Mem Limit 262144 kB

A new pack of n t-shirts came to a shop. Each of the t-shirts is characterized by three integers p_i , a_i and b_i , where p_i is the price of the i -th t-shirt, a_i is front color of the i -th t-shirt and b_i is back color of the i -th t-shirt. All values p_i are distinct, and values a_i and b_i are integers from 1 to 3.

m buyers will come to the shop. Each of them wants to buy exactly one t-shirt. For the j -th buyer we know his favorite color c_j .

A buyer agrees to buy a t-shirt, if at least one side (front or back) is painted in his favorite color. Among all t-shirts that have colors acceptable to this buyer he will choose the cheapest one. If there are no such t-shirts, the buyer won't buy anything. Assume that the buyers come one by one, and each buyer is served only after the previous one is served.

You are to compute the prices each buyer will pay for t-shirts.

Input

The first line contains single integer n ($1 \leq n \leq 200\,000$) — the number of t-shirts.

The following line contains sequence of integers $p_1, p_2, \dots, p_n$ ($1 \leq p_i \leq 1\,000\,000\,000$), where p_i equals to the price of the i -th t-shirt.

The following line contains sequence of integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 3$), where a_i equals to the front color of the i -th t-shirt.

The following line contains sequence of integers $b_1, b_2, \dots, b_n$ ($1 \leq b_i \leq 3$), where b_i equals to the back color of the i -th t-shirt.

The next line contains single integer m ($1 \leq m \leq 200\,000$) — the number of buyers.

The following line contains sequence $c_1, c_2, \dots, c_m$ ($1 \leq c_j \leq 3$), where c_j equals to the favorite color of the j -th buyer. The buyers will come to the shop in the order they are given in the input. Each buyer is served only after the previous one is served.

Output

Print to the first line m integers — the j -th integer should be equal to the price of the t-shirt which the j -th buyer will buy. If the j -th buyer won't buy anything, print -1.

Examples

Input	Output
5 300 200 400 500 911 1 2 1 2 3 2 1 3 2 1 6 2 3 1 2 1 1	200 400 300 500 911 -1

Input	Output
2 1000000000 1 1 1 1 2 2 2 1	1 1000000000

Problem Z. Boxes Packing

Time Limit 1000 ms

Mem Limit 262144 kB

Mishka has got n empty boxes. For every $i (1 \leq i \leq n)$, i -th box is a cube with side length a_i .

Mishka can put a box i into another box j if the following conditions are met:

- i -th box is not put into another box;
- j -th box doesn't contain any other boxes;
- box i is smaller than box j ($a_i < a_j$).

Mishka can put boxes into each other an arbitrary number of times. He wants to minimize the number of *visible* boxes. A box is called *visible* iff it is not put into some another box.

Help Mishka to determine the minimum possible number of *visible* boxes!

Input

The first line contains one integer $n (1 \leq n \leq 5000)$ — the number of boxes Mishka has got.

The second line contains n integers $a_1, a_2, \dots, a_n (1 \leq a_i \leq 10^9)$, where a_i is the side length of i -th box.

Output

Print the minimum possible number of *visible* boxes.

Examples

Input	Output
3 1 2 3	1

Input	Output
4 4 2 4 3	2

Note

In the first example it is possible to put box 1 into box 2, and 2 into 3.

In the second example Mishka can put box 2 into box 3, and box 4 into box 1.